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ECON 412 International Economics: Trade

Trimester 2, 2018

Lecture Week 2: Preliminaries

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Overview

- **Readings:** Feenstra Chapter 1, KOM Chapter 3.
- Refresh microeconomics: producer theory.
- Review of the basic Ricardian model.
- Setup the basics for the Heckscher-Ohlin model.
- Introduction of the Gravity Equation.

-1. Real functions

Differentiation

- $\frac{dg(f(x))}{dx} = \frac{dg}{df} \cdot \frac{df}{dx}$.
- Let $h(x) = f(x) \cdot g(x)$, then $\frac{dh(x)}{dx} = \frac{df(x)}{dx} g(x) + f(x) \frac{dg(x)}{dx}$.
- Let $h(x) = f(x)/g(x)$, then $\frac{dh(x)}{dx} = \left[\frac{df(x)}{dx} g(x) - f(x) \frac{dg(x)}{dx} \right] / g^2(x)$.

Homogeneous functions

- A real function is homogeneous of degree r if $f(tx) = t^r f(x)$ for $t > 0$.
- A function is *homothetic* if it is a positive monotonic transformation of a function of homogeneous of degree 1.
  - E.g., $f(x) = x_1^{\alpha} x_2^{1-\alpha}$ is HOD 1, $g(x) = \ln[f(x)]$ is homothetic.
- **Euler's Theorem:** If $f(x) : \mathbb{R}^n \to \mathbb{R}$ is a differentiable function of homogeneous of degree r, then at any x,
$$rf(x) = \sum_{i=1}^{n} \frac{\partial f(x)}{\partial x_i} x_i.$$
- If $r = 1$, then
$$f(x) = \sum_{i=1}^{n} \frac{\partial f(x)}{\partial x_i} x_i.$$

0. Review of the producer theory

0-1. Cost function

- Notation: y is output quantity, v is a vector of input factors, w is a vector of factor prices.
- Assume, production function $y = f(v)$ is non-negative, strictly increasing, concave, and homogeneous of degree 1.
- Cost minimization: $C(w, y) = \min_v w \cdot v, \text{ subject to } f(v) \geq y$.

- The **cost function** $C(w, y)$ is (see Varian, 1992)
  - increasing in w. If $w > w'$, then $C(w, y) > C(w', y)$.
  - concave in w, i.e., $C(\theta w + (1-\theta)w', y) \geq \theta C(w, y) + (1-\theta)C(w', y)$ for $\theta \in [0, 1]$.
  - homogeneous of degree 1 in w, i.e., $C(tw, y) = tC(w, y)$ for $t > 0$.
  - continuous in w for $w > 0$.

- **Shephard's lemma:** $\frac{\partial C(w,y)}{\partial w_i} = v_i(w, y)$.
- **Proposition:** If production function exhibits constant returns to scale (e.g., homogeneous of degree 1), the cost function is $C(w, y) = C(w, 1)y$.
  - We write $C(w, 1) = c(w)$, the unit cost. Then $C(w, y) = c(w)y$.
- The conditional factor demand is also proportional to output, e.g., $v(w, y) = a(w)y$.

0-2. Profit function

- Profit maximization: $\pi(p, w) = \max_v [pf(v) - w \cdot v]$.
- **Hotelling's lemma:** $\frac{\partial \pi(p,w)}{\partial p} = y(p, w)$.
- $\frac{\partial \pi(p,w)}{\partial w} = -v(p, w)$.

The lemmas are derived by using the envelope theorem.

- The envelope theorem: Let $G(a) = \max_x g(x, a)$, then at the optimal point $x(a)$, $\frac{\partial G(a)}{\partial a} = \frac{\partial g(x,a)}{\partial a}|_{x=x(a)}$.
- Proof: $\frac{\partial G(a)}{\partial a} = \frac{\partial g(x(a),a)}{\partial a} = \frac{\partial g(x(a),a)}{\partial x} \cdot \frac{\partial x(a)}{\partial a} + \frac{\partial g(x(a),a)}{\partial a}$. The first-order necessary condition for the maximization is $\frac{\partial g(x(a),a)}{\partial x} = 0$. ■

1. Ricardian trade theory: two countries and two goods.

- Trade due to differences in technology: Comparative advantge.
- Two countries: Home and Foreign; two goods, $i = 1, 2$.
- Labor endowment: L at Home and L^* in Foreign.
- Labor is perfectly mobile within country, immobile across countries.
- Linear technology, labor needed per unit of good i: a_i and $a_i^*, \text{ for } i = 1, 2$.
- **Comparative advantage**: $a_1/a_2 < a_1^*/a_2^*$: Relatively more productive to produce good 1 at Home.
  - important assumption in Ricardian theory.

1-1. Autarky (focusing on Home, similar for Foreign)

- Production possibility frontiers (PPFs): $\frac{L}{a_1}$, $\frac{L}{a_2}$.
- Wage=marginal value: $w_i = \frac{p_i}{a_i}$. Wage equalization across industries requires: $\frac{p_1}{a_1} = \frac{p_2}{a_2}$.
- Let good 2 be the numeraire ($p_2 = 1$), relative price of good 1 is $p = p_1/p_2$, then $p = a_1/a_2$.
- Real wage (w/p_2) is then given by $\tilde{w} = \frac{p}{a_1} = \frac{1}{a_2}$.
- Similar, in Foreign, relative price is $p^* = a_1^*/a_2^*$. Real wage is $\tilde{w}^* = \frac{p^*}{a_1^*} = \frac{1}{a_2^*}$.
- In autarky, $p < p^*$: relative price is lower at Home.

1-2. Free Trade: want to find equilibrium world price p^t.

- Assumed that labor is immobile between countries.
- Supply:
  - If $p^t < a_1/a_2$, both countries produce good 2 (higher wage in sector 2).
  - If $p^t \in (a_1/a_2, a_1^*/a_2^*)$, Home produces good 1, Foreign produces good 2. World supply: $\frac{L}{a_1}$ and $\frac{L^*}{a_2^*}$.
  - If $p^t > a_1^*/a_2^*$, both countries produce good 1.

- If preferences are homothetic and identical across countries, then the relative demand d_1/d_2 decreases with p^t in either country and in the world.
    - That is, world demand depends only on world income and relative price.
- Equilibrium: world demand equals world supply: giving the equilibrium price p^t. See figure on white board.
- Production:
    - Home: $(L/a_1, 0)$ under free trade; $(L_1/a_1, L_2/a_2)$ under autarky.
    - Foreign: $(0, L^*/a_2^*)$ under free trade; $(L_1^*/a_1^*, L_2^*/a_2^*)$ under autarky.

- *Ricardian Theory*: Trade patterns are determined by comparative advantage, not absolute advantage.

- How do real wages change?

  - At Home, $\tilde{w}^t = \frac{p^t}{a_1} > \frac{p}{a_1} = \tilde{w}$, since $p^t > p$.
  - In Foreign, $\tilde{w}^{*t} = \frac{1}{a_2^*} > \frac{p^t}{a_1^*}$, since $p^t < p^*$.
- Which country has a higher wage under free trade?
  - Depends on absolute advantage.
  - If $a_1 > a_1^*$ (hence $a_2 >> a_2^*$), then $\tilde{w}^t = \frac{p^t}{a_1} < \frac{p^t}{a_1^*} < \tilde{w}^{*t}$.
- Trade is determined by comparative advantage, while wage difference across countries is determined by absolute advantage.

2. Factor endowment: two-good two-factor model

- Set up environment for the Heckscher-Ohlin model. Focus on one country.
- Trade due to differences in factor endowment.
- **Goal is to solve for factor prices and factor allocation, given factor endowment and output prices.**
- **Production technology** to produce good i: $y_i = f_i(L_i, K_i)$, for $i = 1, 2$.

- Assumption: $f_i()$ is increasing and concave in each argument, and homogeneous of degree 1 in (L_i, K_i), implying *constant returns to scale*.

  - will examine increasing returns to scale later.

- Labor and capital are fully mobile between the two sectors. Endowments L and K are given and fixed.

- Want to derive the Production Possibility Frontiers (PPFs). Done by the following problem,

- The maximization problem
$$\max_{L_2, K_2} f_2(L_2, K_2)$$
s.t.
$$\begin{aligned} y_1 &= f_1(L_1, K_1), \\ L_1 + L_2 &\leq L, \\ K_1 + K_2 &\leq K. \end{aligned}$$
The last two inequalities are resource constraints. Solving this problem leads to $y_2 = h(y_1, L, K)$, the **production possibility frontier**.

- **Property**: Function $h()$ is concave in y_1.

2-1. Optimal outputs

- The economy produces at what mix of y_1 and y_2, given p_1 and p_2?
- GDP function, given exogenous output prices p_1 and p_2
  - $G(p_1, p_2, L, K) = \max_{y_1, y_2}(p_1 y_1 + p_2 y_2)$, s.t. $y_2 = h(y_1, L, K)$.
  - First-order necessary condition: $p = \frac{p_1}{p_2} = -\frac{\partial y_2}{\partial y_1}$.
  - In words, the economy will produce (y_1, y_2) at a point where the relative price p equals the (negative) slope of PPF curve.
  - Property: $\frac{\partial G}{\partial p_i} = y_i$. Obained by the envelope theorem (note that $p_1 dy_1 + p_2 dy_2 = 0$ by FONC).

2-2. Equilibrium factor prices (w, r)

- Want to minimize the cost function, conditional on output and given the endowments.
- Cost minimization given the amount of output y_i:
  $C_i(w, r, y_i) = \min_{L_i, K_i} wL_i + rK_i$, s.t. $f_i(L_i, K_i) \geq y_i$.
- **Claim**: If production function $f_i()$ exhibits constant returns to scale, cost function $C_i(w, r, y) = c_i(w, r) \cdot y_i$.
- Thus $c_i(w, r)$ is the cost producing per unit of good i.

- Also useful to recall the Shephard's lemma:
  - $\frac{\partial C_i(w,r,y_i)}{\partial w} = L_i(w,r,y_i) = l_i(w,r)y_i$; the last equality due to CRS.
  - $\frac{\partial C_i(w,r,y_i)}{\partial r} = K_i(w,r,y_i) = k_i(w,r)y_i$.
  - $l_i()$ and $k_i()$ are conditional input demand functions.

- **Equilibrium factor prices** (w, r) are solved by the zero-profit conditions:
$$p_i = c_i(w, r), i = 1, 2.$$
Solution is unique if $c_1(w, r)$ and $c_2(w, r)$ have one common point (e.g., they are single-crossing or tangent).
    - Why is profit zero? Free entry under perfect competition.
- **Lemma**: As long as both goods are produced and unit cost functions have one common point, factor price are then uniquely determined by output prices p_1 and p_2.
- Using resource constraint conditions, we can obtain the **feasible output** y_1 and y_2:
    - $l_1(w, r)y_1 + l_2(w, r)y_2 = L$ and $k_1(w, r)y_1 + k_2(w, r)y_2 = K$.
- There are four equilibrium conditions (2 on zero profit, 2 on resource constraints), and four unknowns (w, r, y_1, y_2), given p_1, p_2, L, and K.
- From the four equations, we can solve for w, r, y_1, y_2.

2-3. Equilibrium outputs

- There are four equilibrium conditions (2 on zero profit, 2 on resource constraints), and four unknowns (w, r, y_1, y_2), given p_1, p_2, L, and K.
- From the four equations, we can solve for w, r, y_1, y_2.
- Solution does not necessarily exist.
- For solution to exist, it requires 1) both goods are produced; 2) unit cost functions cross once or are tangent.
    - 2) says there is no factor intensity reversal (FIR).

3. Trade, factor prices and factor allocation

- How does free trade affect factor prices?
- How does relative output price affect factor use?
- How do changes in factor endowments affect outputs?
- Assumptions:
  - Two goods.
  - Two countries with identical production technology,
  - but different factor endowments.

3-1. Factor price equalization (FPE) Theorem

Suppose that two countries engaged in free trade have identical technologies but different factor endowments. Factors are immobile. If both countries produce both goods and unit cost functions of producing them have one common point, then factor prices are equalized across the countries.

Intuition: Under constant returns to scale, zero-profit condition holds. This together with the one-common-point assumption of unit-cost functions leads to unique mappings between output prices (p_1, p_2) and factor prices (w, r). In the world with free trade, output price of each good equalizes across countries. Therefore factor prices equalize across countries.

Suppose country 1 is labor abundant (relative to country 2), it then can produce and export more of the good that is labor intensive, maintaining the same wage rate as in country 2.

Next, we have two theorems: one deals with the change in output prices, the other deals with changes in endowments.

3-2. Stolper-Samuelson Theorem

> *An increase in the relative price of a good will increase the real return to the factor used intensively to produce that good, and reduce the real return to the other factor.*

Suppose that producing good 1 is labor intensive. If $p = p_1/p_2$ increases, then w/p_1 and w/p_2 will rise.

Proof of Stolper-Samuelson Theorem. The share of labor in total factor cost to produce good i is $\theta_{il} = \frac{wl_i(w,r)y_i}{c_i(w,r)y_i} = \frac{wl_i(w,r)}{c_i(w,r)}$, the share of capital is then $\theta_{ik} = 1 - \theta_{il}$. Zero-profit condition is given by

$$p_i = c_i(w,r) = wl_i(w,r) + rk_i(w,r).$$

The second equality is by definition and also implied by the Euler's Theorem on homogeneous functions.

Noting that $\partial c_i(w,r)/\partial w = l_i(w,r)$ and $\partial c_i(w,r)/\partial r = k_i(w,r)$ by Shephard's lemma, totally differentiating the zero-profit condition leads to:
$dp_i = \frac{\partial c_i(w,r)}{\partial w}dw + \frac{\partial c_i(w,r)}{\partial r}dr = l_i(w,r)dw + k_i(w,r)dr$. Divide both sides by p_i (which equals $c_i(w,r)$), and let $\hat{p}_i = \frac{dp_i}{p_i}$, the percentage change. Then

$$\hat{p}_i = \frac{wl_i(w,r)}{c_i(w,r)}\widehat{w} + \frac{rk_i(w,r)}{c_i(w,r)}\hat{r}.$$

Or

$$\hat{p}_i = \theta_{il}\widehat{w} + \theta_{ik}\hat{r}.$$

In matrix form, we have

$$\begin{bmatrix} \hat{p}_1 \\ \hat{p}_2 \end{bmatrix} = \begin{bmatrix} \theta_{1l} & \theta_{1k} \\ \theta_{2l} & \theta_{2k} \end{bmatrix} \begin{bmatrix} \widehat{w} \\ \hat{r} \end{bmatrix}.$$

The above system of equations can be re-written as

$$\begin{bmatrix} \widehat{w} \\ \hat{r} \end{bmatrix} = \frac{1}{|\theta|} \begin{bmatrix} \theta_{2k} & -\theta_{1k} \\ -\theta_{2l} & \theta_{1l} \end{bmatrix} \begin{bmatrix} \hat{p}_1 \\ \hat{p}_2 \end{bmatrix}$$

where $|\theta| = \theta_{1l} - \theta_{2l} = \theta_{2k} - \theta_{1k}$ is the determinant of the matrix of factor shares.

Suppose that sector 1 is relatively more labor intensive than sector 2, i.e., $\theta_{1l} - \theta_{2l} > 0$. Further, suppose that $\hat{p}_1 - \hat{p}_2 > 0$, meaning that the price of good 1 increases at a faster rate than that of good 2. The wage growth rate is

$\widehat{w} = \frac{1}{|\theta|}(\theta_{2k}\hat{p}_1 - \theta_{1k}\hat{p}_2) = \frac{(\theta_{2k}-\theta_{1k})\hat{p}_1+\theta_{1k}(\hat{p}_1-\hat{p}_2)}{\theta_{2k}-\theta_{1k}} > \hat{p}_1.$

Growth rate of capital rental price is
$\hat{r} = \frac{1}{|\theta|}(-\theta_{2l}\hat{p}_1 + \theta_{1l}\hat{p}_2) = \frac{(\theta_{1l}-\theta_{2l})\hat{p}_2 - \theta_{2l}(\hat{p}_1 - \hat{p}_2)}{\theta_{1l}-\theta_{2l}} < \hat{p}_2.$

Then

$$\widehat{w} > \hat{p}_1 > \hat{p}_2 > \hat{r}.$$

The percentage increase of wage is faster than the percentage inrease in capital rental rate. This means that both w/p_1 and w/p_2 go up. ■

Implication: If relative output price in the world changes, there will be gainers and losers.

3-3. Rybczynski Theorem

> *An increase in a factor endowment will increase the output of the industry using it intensively, and decrease the output of the other industry.*

Suppose that good 1 is labor intensive (relative to good 2) and let's fix output prices at some values, then an increase in labor L leads to an increase in y_1 and a decrease in y_2.

Proof. Recall that the optimal conditional input demands for labor and capital in producing good i are respectively, $l_i(w,r)y_i$ and $k_i(w,r)y_i$. They are proportional to a given amount of output, a result owing to the constant resturns to scale of production function. The resource constraints are then

$$L = l_1(w,r)y_1 + l_2(w,r)y_2$$
$$K = k_1(w,r)y_1 + k_2(w,r)y_2.$$

Total differentiating the equations, and after some manipulation, we have

$$\frac{dL}{L} = \frac{l_1(w,r)y_1}{L}\frac{dy_1}{y_1} + \frac{l_2(w,r)y_2}{L}\frac{dy_2}{y_2}$$
$$\frac{dK}{K} = \frac{k_1(w,r)y_1}{K}\frac{dy_1}{y_1} + \frac{k_2(w,r)y_2}{K}\frac{dy_2}{y_2}.$$

NOTE: p_1 and p_2 are fixed at some values, thus so are w and r.

Let $\lambda_{il} = \frac{l_i(w,r)y_i}{L}$ and $\lambda_{ik} = \frac{k_i(w,r)y_i}{K}$, for $i = 1, 2$. Also, denote $dx/x = \widehat{x}$. The above system of two equations is written in matrix form as

$$\begin{bmatrix} \widehat{L} \\ \widehat{K} \end{bmatrix} = \begin{bmatrix} \lambda_{1l} & \lambda_{2l} \\ \lambda_{1k} & \lambda_{2k} \end{bmatrix} \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \end{bmatrix}.$$

Re-arranging the coefficients, we obtain

$$\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \end{bmatrix} = \frac{1}{|\lambda|} \begin{bmatrix} \lambda_{2k} & -\lambda_{2l} \\ -\lambda_{1k} & \lambda_{1l} \end{bmatrix} \begin{bmatrix} \widehat{L} \\ \widehat{K} \end{bmatrix}.$$

Here $|\lambda|$ is the determinant of the coefficient matrix, $|\lambda| = \lambda_{1l} - \lambda_{1k} = \lambda_{2k} - \lambda_{2l}$. Here, note that $\lambda_{1l} + \lambda_{2l} = 1$ and $\lambda_{1k} + \lambda_{2k} = 1$).

Suppose that sector 1 (good 1) is labor intensive, i.e., $\lambda_{1l} > \lambda_{1k}$ (implying $\lambda_{2l} < \lambda_{2k}$). Then $|\lambda| > 0$. We have

$$\hat{y}_1 = \frac{\lambda_{2k}\widehat{L} - \lambda_{2l}\widehat{K}}{\lambda_{2k} - \lambda_{2l}};$$
$$\hat{y}_2 = \frac{-\lambda_{1k}\widehat{L} + \lambda_{1l}\widehat{K}}{\lambda_{1l} - \lambda_{1k}}.$$

Thought experiment: Let $\widehat{K} = 0$ (fix capital level unchanged), and $\widehat{L} > 0$ (labor endownment increases). It is straightforward that $\hat{y}_1 > 0$ and $\hat{y}_2 < 0$. ■

NOTE: $\lambda_{1l} > \lambda_{1k}$ is equivalent to $\frac{L_1}{K_1} > \frac{L}{K} > \frac{L_2}{K_2}$.

Remarks

Relative price $p = p_1/p_2$ is exogenous in the above theorems because they are about the producers' behavior. Price p can be solved by bringing the household's optimization problem, and setting demand equal to supply for good 1.

In the world with free trade, p is determined by the world supply and world demand for good 1 (all relative to good 2).

4. The Gravity Equation

- The Gravity Equation describes a relation between the trade flow and (income, trade barrier).
- Gravity Equation has several variations, below is one of them

$$X_{ni} = \frac{Y_i}{\Omega_i} \cdot \frac{Y_n}{\Phi_n} \cdot \frac{1}{d_{ni}}.$$

  - Here, X_{ni} is the bilateral trade flow between country n and country i. Y_i and Y_n are value of production, respectively, in country i and country n. Ω_i and Φ_n are 'multilateral resistance', and d_{ni} is the trade barrier (which includes but is not limited to geographical distance between n and i).
- Major new trade models all lead to some form of the Gravity Equation, we will study some of them.